

# The fancyqr package



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Version v2.4 – 2026/08/17

LPPL 1.3c or later

[github.com/EagleoutIce/fancyqr](https://github.com/EagleoutIce/fancyqr)

fancyqr is a simple package to create fancy qr-codes with the help of the *qrcode*-package. You can use the `\fancyqr`-macro as a drop-in replacement for the normal `\qrcode`.<sup>1</sup>

If you do want to hide a center square (e.g. because you want to embed an image) you can use `\FancyQrDoNotPrintSquare{<x>}{<y>}` to hide a rectangle with radius *x* and *y* set from the center (with `\FancyQrDoNotPrintRadius{<factor>}` you can apply a *rounding* to this!). If you choose this option, the default `\FancyQrRoundCut` that rounds cut corners can be changed with `\FancyQrHardCut`. The styles that draw randomized tiles use the random number generator of  $\text{\LaTeX}$ ; `\FancyQrSeed{<n>}` (or the seed option) fixes its seed and makes them reproducible.

All of the extra qr-options (you can set all of them with `\fancyqrset{<keys>}`) are showcased in Table 1. The defaults are set like this:

```
\fancyqrset{image padding=0,gradient=true,
            gradient angle=135,
            r color=red!68!black!88!white!90!white,
            l color=purple!40!red!20!black!90!white}
```

Consider the following examples (uses *fontawesome*, but you can use include images, ...):

```
\fancyqr[
  classic,image=\LARGE\faGithub,
]{https://github.com/EagleoutIce/fancyqr}
```



```
\FancyQrLoad{goo}%
\fancyqr[
  seed=3,content color=black
]{https://github.com/EagleoutIce/fancyqr}%
```



```
% needs \usepackage[prefix=@]{xcolor-material}
\FancyQrLoad{dots}%
\fancyqr[
  height=4cm,
  image={\textbf{\textcolor{gray}{fancy}qr}},
  random color={@Red800}{@Purple800}{@Indigo
    800}{@Teal800}{@Green800}
    {@DeepOrange800}{@Brown700}{@BlueGrey
    700}
]{https://github.com/EagleoutIce/fancyqr}%
```



<sup>1</sup>`\fancyqr[<qr-options>]{<url>}`

# 1 Styles

Besides the `default` style there are eight more, all loaded (locally) with `\FancyQrLoad{<name>}`; `\FancyQrLoad{default}` and `\FancyQrLoadDefault` reset to the default. Figure 1 shows every one of them with the same payload and the call that produced it.



Figure 1: Every style that `fancyqr` ships with. `blobs`, `glitch` and `goo` are randomized and therefore shown with a fixed seed. Every one of them but `frame` is read by a scanner as it is; `frame` leaves the center of a module white, which is where a scanner samples, so it needs `finder=square` to be found at all.

The `default` style rounds the free corners of a tile with the `rounding` radius. A corner that is enclosed by two black neighbours but not by the diagonal one is an *inner* corner; it is softened with a fillet of the (smaller) `inner rounding` radius instead of staying a hard 90 degree edge. Setting `inner rounding=0`, or loading the `rounded` style, brings back the sharp notches of earlier versions:

```
\fancyqr[
  color=black,inner rounding=0
]{https://github.com/EagleoutIce/fancyqr}
```



```
\fancyqr[
  color=black,rounding=.25,inner rounding=.35
]{https://github.com/EagleoutIce/fancyqr}
```

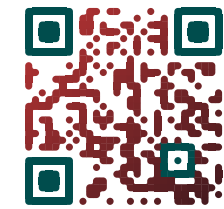


Every style picks the pattern shape that suits it: `dots` and `blobs` a circle, everything else `inherit`, which leaves the patterns to the tiles of the style itself. The `glows` style draws every module once per halo ring and once for the tile itself; `\FancyQrPasses{<n>}` is what a style uses to ask for that. It is tuned with `\FancyQrGlow{<width>}{<strength>}{<falloff>}` – the width of the widest ring in modules, the percentage of color it keeps, and how quickly the rings gain color towards

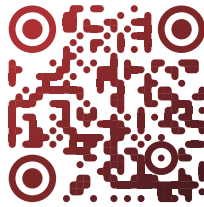
the tile – and with `\FancyQrGlowColor{<from>}{<to>}`, where `.` is the color of the tile the halo surrounds: `\FancyQrGlowColor{white}{black}` turns it into a white backglow. Keep the width below one module, or the halos of two modules with a white one between them run into each other. `glitch` uses the passes as well, for the two displaced ghosts it puts behind its shards; its `\fancyqr@glitch@detach` is the chance that a shard tears off a neighbour instead of merging with it. `goo` hangs its tiles downwards by `\fancyqr@goo@drip`; `0` makes it symmetric again. `frame` outlines the shape the modules form – only the sides facing white are stroked, so a run of modules becomes one pill – and puts a dot inside every module, because a scanner samples the center of a module and a bare outline would leave it white. `dots` and `blobs` additionally draw the position and alignment patterns as solid shapes, because a scanner first has to find those; with `finder=inherit` they are made of tiles as well and the code no longer reads. A style declares with `\FancyQrInnerRoundings` (or `\FancyQrNoInnerRoundings`) whether its tiles understand inner corners at all; only `default` and `glows` do, so no other style pays for the extra lookups.

## 2 Position and alignment patterns

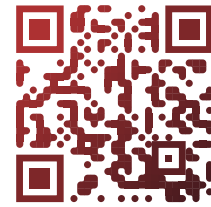
The three big position (finder) patterns and all alignment patterns can be drawn as one shape instead of as single modules, and in a color of their own. `finder` picks the shape of their ring, `finder_core` the shape of their center block (`auto` follows the ring), `finder_radius` and `finder_thickness` its corner radius and its ring width, and `finder_color` paints both. **Figure 2** shows the combinations together with the keys that produced them. `finder_color` paints the patterns and `content_color` everything else, so `content_color=black` keeps the gradient on the patterns alone. The modules of a pattern stay black in the matrix and only their drawing is replaced, so neighbouring tiles still merge into the shape wherever it allows it. A `finder_thickness` other than `1` leaves the `1:1:3:1:1` run a scanner looks for, so check that the code still reads. `finder=inherit` is the default and leaves the patterns to the tiles of the loaded style, which is what `fancyqr` did before the shapes existed; `none` is accepted as a synonym. A `finder_core` set by name still applies there: the ring keeps its tiles and only the center block becomes a shape, which is often all a code needs to stop looking generic. `finder_core=inherit` hands the core back to the tiles, and leaves it empty when the ring is a shape and there are no tiles left to inherit from. Loading a style resets `finder` and `finder_core`, because a style owns its shapes.



`\fancyqr[finder  
color=teal!60!black]`



`\fancyqr[finder=circle]`



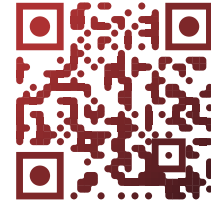
`\fancyqr[finder=square]`



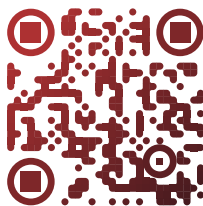
`\fancyqr[finder=rounded]`



`\fancyqr[finder=rounded,  
finder core=star]`



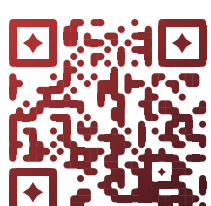
`\fancyqr[finder=square,  
finder core=circle]`



`\fancyqr[finder=circle,  
finder core=square]`



`\fancyqr[finder=rounded,  
finder radius=.15]`



`\fancyqr[finder core=star]`

Figure 2: Styling the position and alignment patterns.

## 3 History

### Version v2.4 2026-08-17

<https://github.com/EagleoutIce/fancyqr/releases/tag/v2.4>

- Weakly rounded (negative) inner corners for the **default** style
- New **rounded**, **glows** and **goo** styles
- Style the position and alignment patterns (**finder**, **finder core**, **finder color**, **finder radius**, **finder thickness**) and color them apart from the data with **content color**
- Configurable **rounding** and **inner rounding**, reproducible randomness with **seed**, a less red default gradient
- Roughly five times faster (the module matrix of a code is kept, see **cache**), an **l3build** test suite
- Tile overlap is relative to the module size (**overlap**), which keeps renderers from leaving seams between the tiles
- Bug fixes: **frame** is readable by a scanner again, the center module is no longer dropped without a cut-out, **gradient angle** is read in degrees again, and an embedded image is centered exactly

### Version v2.3 2026-02-27

<https://github.com/EagleoutIce/fancyqr/releases/tag/v2.3>

- Relicense under the LPPL 1.3c

<sup>2</sup>The package will automatically calculate the required **\FancyQrDoNotPrintSquare** (you have to make sure that the qr-code still has enough information to be readable). Therefore, the image will not scale with the qr-code.

| Option           | Type                            | Default   | Explanation                                                                                                                                                                                                                                                                                                                         |
|------------------|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cache            | boolean                         | true      | Keep the module matrix of a code, so that drawing it again – in a gallery, a running header, ... – only has to redraw it.                                                                                                                                                                                                           |
| classic          | boolean                         | false     | Use the classic qr-code style (black with flat rectangles, this loads the flat style).                                                                                                                                                                                                                                              |
| color            | color                           |           | Disables the gradient and sets the color accordingly.                                                                                                                                                                                                                                                                               |
| compensate       | length                          |           | Overlap between neighbouring tiles as an absolute length; setting it switches off the module-relative overlap.                                                                                                                                                                                                                      |
| finder color     | color                           |           | Color of the position and alignment patterns. Without finder they keep their normal tiles and are only recolored.                                                                                                                                                                                                                   |
| content color    | color                           |           | Color of everything that is not a position or alignment pattern, so that the patterns can keep the gradient while the data stays flat.                                                                                                                                                                                              |
| finder           | shape                           | inherit   | Draw the three position (finder) patterns and all alignment patterns as one shape (square, rounded, or circle) instead of as single modules; inherit (synonym none) leaves them to the tiles of the style (section 2).                                                                                                              |
| finder core      | shape                           | auto      | Shape of the center block of every pattern (auto, inherit, none, square, rounded, circle, or star); auto follows finder, any other value applies even when finder leaves the ring to the tiles (section 2).                                                                                                                         |
| finder radius    | number                          | 0.5       | Corner radius of the rounded pattern shape, relative to the pattern radius.                                                                                                                                                                                                                                                         |
| finder thickness | number                          | 1         | Width of the pattern ring, in modules.                                                                                                                                                                                                                                                                                              |
| gradient         | boolean                         | true      | Toggle the color gradient                                                                                                                                                                                                                                                                                                           |
| gradient angle   | angle                           | 135       | Change the gradient angle, in degrees; 0 runs from left to right.                                                                                                                                                                                                                                                                   |
| image            | L <sup>A</sup> T <sub>E</sub> X |           | Automatically center an image (you have to care for the size and maybe adjust the version and level to keep the qr-code readable). <sup>2</sup>                                                                                                                                                                                     |
| image padding    | number                          |           | Additionally hide blocks (x & y) around the image.                                                                                                                                                                                                                                                                                  |
| image x padding  | number                          | 0         | Additionally hide blocks (x) around the image.                                                                                                                                                                                                                                                                                      |
| image y padding  | number                          | 0         | Additionally hide blocks (y) around the image.                                                                                                                                                                                                                                                                                      |
| inner rounding   | number                          | 0.25      | Radius of the negative rounding: a corner enclosed by two black neighbours but not by the diagonal one gets a fillet of this size, 0 disables it (section 1).                                                                                                                                                                       |
| overlap          | number                          | 0.045     | Overlap between neighbouring tiles, as a fraction of a module. Two tile edges that cross the same pixel each cover only a part of it, and a renderer then leaves a sliver of the background showing through; what decides whether that is visible is how many device pixels the overlap is, which is why it scales with the module. |
| l color          | color                           | see above | Color at the left end of the gradient axis, which the default angle puts at the bottom right.                                                                                                                                                                                                                                       |
| left color level | color<br>L/M/Q/H                | M         | Alias for l color.<br>qrcode option affecting error correction (low, medium, quartile, high).                                                                                                                                                                                                                                       |
| padding          | flag                            |           | qrcode option adding sufficient additional space around the qr-code.                                                                                                                                                                                                                                                                |
| r color          | color                           | see above | Color at the right end of the gradient axis, which the default angle puts at the top left.                                                                                                                                                                                                                                          |
| random color     | colors                          |           | Allow to set a random color pool to pick from.                                                                                                                                                                                                                                                                                      |
| right color      | color                           |           | Alias for r color.                                                                                                                                                                                                                                                                                                                  |
| rounding         | number                          | 0.5       | Corner radius of the rounded tiles of the default style.                                                                                                                                                                                                                                                                            |
| seed             | integer                         |           | Fix the random seed, so that the randomized styles (blobs, glitch, goo) render reproducibly.                                                                                                                                                                                                                                        |
| size             | length                          |           | Alias for qrcode's height option.                                                                                                                                                                                                                                                                                                   |

### **Version v2.2** 2024-11-27

<https://github.com/EagleoutIce/fancyqr/releases/tag/v2.2>

- Configurable tile overlap
- Circular cut-out support
- New `compensate` option

### **Version v2.1** 2024-10-05

<https://github.com/EagleoutIce/fancyqr/releases/tag/v2.1>

- Add `classic` option to easily typeset classic QR codes
- Simplify picture usage for QR codes
- More documentation and Bug-Fixes

### **Version v2.0** 2024-04-13

<https://github.com/EagleoutIce/fancyqr/releases/tag/v2.0>

- No longer require TikZ!
- Random Colors
- Support `width` and `height`

### **Version v1.0** 2022-08-18

<https://github.com/EagleoutIce/fancyqr/releases/tag/v1.0>

- Initial release with basic drop-in functionality
- Using TikZ to create the QR code